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The Effect of Financial Literacy Levels on Gambling Behaviour

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Abstract

This study examines how financial literacy relates to gambling behaviour in a Turkish adult sample. Using a cross-sectional, face-to-face survey (N = 182), we assessed gambling severity (SOGS) and a multidimensional financial literacy scale (FL). Spearman analyses indicated a small, borderline inverse association between FL Total and SOGS ($\rho = -0.140$, $p = .059$), while the Fundamentals of Finance (FF) subscale showed a significant negative correlation ($\rho = -0.188$, $p = .011$). Education and income were associated with differences in outcomes: lower education and lower income groups tended to report higher SOGS and lower FL. Gender and age did not differ on SOGS or FL Total; contrasts appeared only for the Cryptocurrencies/Digital Assets (CDA) subscale (men and younger adults scored higher). In the sample, 40.1% had ever gambled and 2.7% met the pathological threshold. Overall, the correlational pattern points to a modest inverse co-variation between core financial skills and gambling severity in this dataset; causal inferences are not warranted. Limitations include cross-sectional design and self-report measures.

Keywords

Basic finance knowledge • Financial literacy • Gambling behavior

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Introduction

Gambling, despite being a widespread socio-economic phenomenon that has persisted throughout history in various societies, has become more prevalent and accessible with the technological transformations brought about by the digital age. All forms of gambling, defined as investing money or offering something of material value in an activity with an uncertain outcome in order to obtain more money or material value, have the potential to lead to addiction (Williams et al., 2017). The internet increases the risk of gambling addiction by making online gambling more appealing, as it provides easy access to gambling regardless of location or time constraints and allows users to conceal their identity (Griffiths, 2003).

Although the figures regarding the prevalence of gambling in Türkiye are unclear, relevant reports indicate that the problem may be more widespread than previously thought (Öztürk, 2021). A report prepared by the Istanbul Planning Agency using data shared by We Social and Meltwater states that, as of February 2025, the global average for playing virtual gambling or betting in the last month was 7.6%, while in Türkiye this rate was 10.4%. The virtual gambling market in Türkiye is estimated to exceed 1 billion dollars in 2024, with this figure standing at 985.5 million dollars in 2023 (İstanbul Planlama Ajansı, 2025). However, according to Yeşilay's 2023 Activity Report, a total of 13,930 new applications were made to the Yeşilay Counselling Centre in 2023, and 4,230 of these applications were related to gambling addiction. The report, covering the years 2021–2023, reported 9,368 individuals receiving counselling for gambling addiction and 573 group therapy sessions involving 939 gambling addicts (Türkiye Yeşilay Cemiyeti, 2024). These data, provided by national and international institutions, highlight that gambling is an increasingly significant problem.

The American Psychiatric Association (APA) addresses gambling disorder in the DSM-5 under the category of 'substance-related and addictive disorders'; defining the primary feature of gambling disorder as persistent and recurrent gambling behaviour of a destructive nature that negatively impacts the individual's personal, family, or occupational life (American Psychiatric Association, 2022). Psychological factors such as impulsivity, stress, alexithymia, dissociative experiences, depression, anxiety, mood disorders, and substance use disorders, as well as various sociodemographic characteristics, are known to be risk factors in the development and maintenance of gambling disorder (Moreira et al., 2023). Research on gambling indicates relatively consistent results in terms of sociodemographic characteristics, despite some variability; gambling problems are particularly prevalent among males, single or divorced individuals, young people, those with low levels of education, ethnic minorities or those born abroad, the unemployed, and people with low incomes (Calado & Griffiths, 2016). Griffiths (2005) emphasises that biological predispositions, psychological sensitivities and the social context play an important role in the development and maintenance of addictive behaviours. On the other hand, the social harms of gambling are seen in a range of areas, including financial, health, emotional/psychological, family/relationships, work/school, and illegal activities (Volberg et al., 2021).

Financial literacy refers to the knowledge base necessary for an individual to make correct and appropriate decisions regarding investments, debts, and other financial matters (Neşleha, 2017). In other words, it can also be defined as the ability to make informed judgements and effective decisions regarding the use and management of money (Worthington, 2013). Gambling behaviour and financial literacy are suggested to be related in terms of potential causality and consequences (Watanapongvanich et al., 2021; Watanapongvanich et al., 2022).

The definition of gambling addiction in DSM-5 covers all gambling activities, including those on digital platforms. Virtual gambling, opportunities for gambling-themed games and activities are increasing, and it has been demonstrated that virtual activities marketed as ‘games’ by the gambling industry can possess the fundamental structural characteristics of gambling (e.g., chance-based reward mechanisms, betting with virtual money), even if no monetary gain is achieved (King et al., 2015). This situation may lower risk perception, thereby creating conditions conducive to the development of pathology in individuals prone to gambling. However, King et al. (2015) emphasise that the fundamental distinction between gambling and gaming activities lies in whether financial gain or loss is involved. Given that the financial loss aspect of gambling can be devastating and that financial distress is a risk factor for gambling in its own right (Moreira et al., 2023), this distinction is critical. The fact that the financial dimension is the decisive factor in the transition from simulated experiences to risky “real” gambling in hybrid activities (King et al., 2015), that incorporate characteristics of gambling such as virtual money, chance, and betting but cannot be defined as “gambling” either legally or in the field of mental health raises the question of whether financial literacy is a variable related to this process.

It has been revealed that individuals gamble more in response to economic crises and financial difficulties, using games of chance as a shortcut to making money or as a solution to financial problems (Olason et al., 2015). Individuals without a regular income or those in debt may seek solutions to these difficulties through gambling, but as gambling behaviour increases, they may experience greater financial losses (İstanbul Planlama Ajansı, 2025). Increased gambling losses create serious problems for gamblers' savings and investments (Watanapongvanich et al., 2021). Financial literacy is a critical skill for individuals to understand how to manage their money, budgets, debt, and investments (Lusardi & Mitchell, 2013). In this regard, it is stated that educational programmes may contribute to reducing the likelihood of gambling by raising awareness about the financial consequences and risks involved in gambling and by supporting healthier financial habits (Williams et al., 2012).

According to Kahneman and Tversky's (1979) prospect theory, individuals code their financial decisions in terms of ‘gains’ and ‘losses’ using the current situation as a reference point; the pain caused by losses is stronger than the pleasure derived from gains of equivalent magnitude. This asymmetric attitude can trigger the tendency to ‘chase losses’ (loss chasing), commonly seen in gambling, leading individuals to take disproportionately greater risks in the hope of returning to their starting point (Kahneman & Tversky, 1979). On the other hand, financial decisions are often influenced by cognitive shortcuts rather than rational analysis; intuitive processes such as representativeness, availability, and anchoring and adjustment cause systematic and predictable errors based on intuitive judgements rather than statistical reality (Tversky & Kahneman, 1974).

It is known that cognitive distortions are common among pathological gamblers, with the most prominent being the ‘illusion of control,’ defined as the belief that one has more personal control over chance-based events than is actually the case (Goodie & Fortune, 2013). Similarly prevalent is the gambler's fallacy, which is the mistaken belief that after a series of random events, an alternative outcome is ‘due’ (e.g., the thought that ‘it's come up heads five times now, so the probability of tails is very high’), directly triggering loss chasing (Tversky & Kahneman, 1974). In addition to these, overconfidence, self-serving bias, the near-miss effect, and biases related to availability are also significant distortions that perpetuate gambling behaviour (Goodie & Fortune, 2013).

Overweighting low probabilities, particularly in situations where the probability of winning is very low (e.g., national lottery), can make even mathematically disadvantageous gambles attractive. However, individuals tend to mentally categorise money according to its source. For example, gamblers may be more willing to risk money earned from gambling (money from the pot) than money earned through harder means, such as a salary. This reflects a tendency to view money not holistically but in relation to context-sensitive reference points (Kahneman & Tversky, 1979).

Studies examining the relationship between financial literacy and gambling behaviour highlight that those with higher financial literacy gamble less, but emphasise that this relationship may vary depending on access to gambling and cultural characteristics (Barone & Graffigna, 2025). The number of studies examining both variables together in the Turkish sample is quite limited. This research is one of the first studies to examine the relationship between the level of financial literacy and indicators of problem gambling in the Turkish context. Accordingly, the aim is to reveal the relationship between general financial literacy and its sub-dimensions (financial fundamentals, investment instruments and financing, debt and risk management, personal finance management, cryptocurrencies and digital assets, personal financial planning) and gambling behaviour. It also examined whether these variables differ according to sociodemographic variables such as age, education, and income, as well as the relationship between having received financial literacy education in the past and participation in the individual pension system. The study is expected to contribute to the literature as one of the pioneering studies examining the relationship between gambling and financial literacy in the Turkish sample.

Method

This cross-sectional study analyzed survey data collected face-to-face from N = 182 adult participants. Participants were recruited via non-probability convenience sampling, approaching available adults at random. All variables used in the analyses were complete (no missingness). Key sociodemographic distributions (e.g., gender, age bands, education, income, savings behavior, marital status, individual pension system (IPS; Turkish: BES) participation, prior financial literacy training, and internet banking use) are summarized in Table 1; gambling behavior descriptors are in Table 2.

Table 1

Sociodemographic characteristics of the sample (N = 182)

Characteristic	Category	n	%
Gender	Female	58	31.9
	Male	124	68.1
Age	20–29	57	31.3
	30–39	51	28.0
	40–49	51	28.0
	50+	23	12.6
Marital status	Married	108	59.3
	Single/Divorced	74	40.7
Education level	Primary/Secondary	32	17.6
	Bachelor's	109	59.9
	Postgraduate	41	22.5
Income level	Low	68	37.4
	Middle	102	56.0
	High	12	6.6
Savings rate	None	46	25.3
	0–5%	47	25.8
	6–9%	24	13.2
	10–14%	18	9.9
	15–19%	47	25.8
IPS (individual pension) participation	Yes	52	28.6
	No	130	71.4
Financial literacy training (ever)	Yes	18	9.9
	No	164	90.1
Internet banking use	Yes	182	100.0
	No	0	0.0

Table 2

Gambling behavior descriptors (N = 182)

Panel A. Gambling activity types by frequency			
Activity type	Never	< weekly	≥ weekly
Horse-race betting	181	1	0
Card games for money (e.g., Yanık, poker)	179	2	1
Okey for money (tile game)	168	12	2
Dice games for money (e.g., barbut)	181	1	0
Cockfighting	179	2	1
Sports Toto or Sports Lotto (Sports betting pools)	156	22	4
Lottery (Sayısal Loto)	160	20	2
Scratch-off	160	22	0
National Lottery	146	35	1
Stock-market speculation	143	31	8
Casino games	173	6	1
Playing skill-based games for money	176	5	1
Other types of gambling	181	0	1
Panel B. Ever gambled at least once			
Response	n	%	
Yes	73	40.1	
No	109	59.9	
Total	182	100.0	
Panel C. Ever deposited money for gambling at least once			Min-Max
Yes	49	26.9	5-1.000.000
No	133	73.1	
Total	182	100	
Panel D. Pathological gambling			Median (IQR)
Yes	5	2.7	
No	177	97.3	
Total	182	100.0	

Note. IQR = interquartile range.

Ethical Approval

Ethical approval for this study was obtained from Necmettin Erbakan University ethics committee (Decision No: 2025/5575, Date: 21.02.2025), and the study was supported by the university's Scientific Research Projects program. Informed consent was obtained from all participants.

Measures

Two instruments were used: the South Oaks Gambling Screen (SOGS) and the Financial Literacy scale (FL). Additionally, we collected sociodemographic information (gender, age bands, education, income, savings behavior, marital status, individual pension system (IPS) participation, prior financial literacy training, and internet banking use). Throughout the manuscript we use the following abbreviations: SOGS (South Oaks Gambling Screen), FL Total (Financial Literacy total score), FF (Fundamentals of Finance), IIF (Investment Instruments and Financing), DRM (Debt and Risk Management), PFM (Personal Finance Management), CDA (Cryptocurrencies and Digital Assets), and PFP (Personal Financial Planning).

South Oaks Gambling Screen (SOGS). We used the validated Turkish form of SOGS ([Lesieur & Blume, 1987](#); [Turkish version: Duvarcı & Varan, 2001](#)). The Turkish form is scored on 19 items with a 0–19 total range; higher scores indicate greater gambling problems, and a commonly used threshold of 8 indicates probable pathological gambling. Items are answered in mainly dichotomous (Yes/No) with a few simple multiple-choice responses; per the Turkish scoring guide, the first three context items are not scored and risk-indicating responses on subsequent items are summed to obtain the total.

Financial Literacy Scale (FL). The FL scale ([Özyeşil & Tembelo, 2024](#)) comprises 32 items collected on a 5-point Likert format and is organized into six subscales: FF, IIF, DRM, PFM, CDA, and PFP. Total and subscale scores were computed (higher scores reflect greater financial literacy).

Data Analysis

All analyses were conducted in R 4.5.0 ([R Core Team, 2025](#)). Packages used in the workflow included `foreign`, `haven`, `dplyr`, `car`, `carData`, `nortest`, `effectsize` (plus base stats functions). Data completeness was verified (no missing values). Distributional diagnostics indicated pronounced non-normality and zero-inflation for SOGS and mild-to-moderate deviations from normality for several financial literacy variables. Normality was assessed via Shapiro–Wilk test together with histograms and Q–Q plots; variance homogeneity was assessed via Levene’s test.

Given these diagnostics, for two-group comparisons we used non-parametric tests (Mann–Whitney U) as the primary approach for all outcomes. Multi-group contrasts (age, education, income, savings) used Kruskal–Wallis with Dunn post-hoc pairwise comparisons; multiplicity was controlled with Bonferroni. Correlations between SOGS and FL Total/subscales used Spearman’s ρ . Where reported, 95% confidence intervals for ρ were obtained via nonparametric bootstrap (percentile CIs). For Mann–Whitney U we report Glass rank-biserial correlation (r_{rb}) with 95% CIs; for Kruskal–Wallis we report epsilon-squared (ϵ^2).

For two-group comparisons that showed only mild departures from normality we additionally ran parametric t-tests (Student/Welch) as sensitivity checks. All inferential decisions were 100% concordant with the non-parametric results (identical significance and direction). The significance threshold was $\alpha = .05$ (two-sided), and

post-hoc pairwise comparisons were Bonferroni-adjusted. For all analyses, the sex variable was handled and reported as gender.

Findings

Sample Characteristics and Gambling Context

The sample comprised 58 women (31.9%) and 124 men (68.1%); age groups were 20–29 (31.3%), 30–39 (28.0%), 40–49 (28.0%), and 50+ (12.6%). Education levels were lower/upper secondary (17.6%), bachelor's (59.9%), and postgraduate (22.5%). Income groups were low (37.4%), middle (56.0%), and high (6.6%). Savings categories ranged from “none” (25.3%) to 15–19% of income (25.8%). Ever gambled at least once: 40.1%, ever deposited money for gambling at least once: 26.9%, and pathological gamblers: 2.7%. Median self-reported maximum amount ever deposited for gambling in a single day was 500 (IQR = 4,850; min = 5; max = 1,000,000; local currency). Internet banking use was 100%. Descriptive statistics for SOGS and all financial literacy indices are reported in Table 3.

Table 3

Descriptive statistics for gambling and financial literacy scales (N = 182)

Scale	N	Mean	SD	Median	IQR	Min	Max
SOGS	182	0.78	2.23	0	1	0	17
FL Total	182	109.92	13.49	109	17	67	152
FF	182	29.98	4.56	29.50	6	19	43
IIF	182	16.22	2.42	16	2.39	9	25
DRM	182	24.10	4.14	24	4	11	53
PFM	182	18.40	3.42	18.50	4	5	25
CDA	182	14.06	2.93	14	4	8	20
PFP	182	7.16	1.45	7	2	2	10

Note. SD = standard deviation; IQR = Q3 – Q1. SOGS = South Oaks Gambling Screen, FL Total = Financial Literacy total score, FF = Fundamentals of Finance, IIF = Investment Instruments and Financing, DRM = Debt and Risk Management, PFM = Personal Finance Management, CDA = Cryptocurrencies and Digital Assets, and PFP = Personal Financial Planning.

Bivariate Associations Between Financial Literacy and Gambling

The association between overall financial literacy and gambling was small and borderline non-significant: FL Total–SOGS $\rho = -0.140$; $p = .0587$ (95% bootstrap CI $[-0.295, 0.021]$). The Fundamentals of Finance subdimension showed a significant negative association with gambling: FF–SOGS $\rho = -0.188$; $p = .011$ (95% bootstrap CI $[-0.336, -0.019]$). For scale cohesion, FL Total correlated strongly/very strongly with its subscales and related indices—for example: FL Total–DRM $\rho = .813$ (95% bootstrap CI $[.740, .867]$), FL Total–PFM $\rho = .791$ (95% bootstrap CI $[.726, .846]$), FL Total–CDA $\rho = .771$ (95% bootstrap CI $[.690, .832]$); all $p < .001$. Most remaining subscale-to-subscale correlations were moderate to strong and statistically significant. Spearman correlations between gambling and financial literacy scores are presented in Table 4.

Table 4

Spearman Correlation matrix between gambling and financial literacy scores (N = 182)

	SOGS	FL Total	FF	IIF	DRM	PFM	CDA	PFP
SOGS	—	0.059	0.011	0.530	0.081	0.247	0.152	0.932
FL Total	-0.140	—	< .001	< .001	< .001	< .001	< .001	< .001
FF	-0.188*	0.643**	—	0.025	< .001	< .001	< .001	0.046
IIF	-0.047	0.596**	0.166*	—	< .001	< .001	< .001	0.029
DRM	-0.130	0.813**	0.321**	0.514**	—	< .001	< .001	< .001
PFM	-0.086	0.791**	0.361**	0.465**	0.590**	—	< .001	< .001
CDA	-0.107	0.771**	0.475**	0.420**	0.593**	0.513**	—	< .001
PFP	0.006	0.475**	0.148*	0.162*	0.402**	0.454**	0.307**	—

Note. Below the diagonal: Spearman's ρ (* $p < .05$; ** $p < .001$). Above the diagonal: two-sided p-values ($< .001$ shown explicitly).

Group Differences

In the two-group contrasts, only CDA differed by gender ($U = 2942.5$, $p = .044$, $r_{rb} = -0.182$, 95% CI [-0.349, -0.003]), with men scoring higher than women (14.42 vs 13.31). No gender differences emerged for SOGS, FL Total, or the other subscales, and neither marital status, IPS participation, nor prior financial literacy training produced significant contrasts (all $p \geq .05$; for FL Total by prior training, $p = .072$). Two-group comparisons by gender, marital status, IPS participation, and prior financial literacy training are shown in Table 5.

Table 5

Two-group comparisons (Mann-Whitney U): Gender, marital status, ips participation, and prior financial literacy training (N = 182)

Gender										
Female (n = 58)				Male (n = 124)			U	Z	p	r _{rb}
Scale	Median	IQR	Mean R.	Median	IQR	Mean R.				
SOGS	0	0	85.2	0	1	94.4	3233.0	-1.09	.16	-.10
FL Total	108	15	85.6	107.9	19	94.3	3255.0	-1.03	.30	-.09
FF	30	6	94.8	29	6.3	89.9	3402.0	-0.59	.56	.05
IIF	15	2	82.4	16	3	95.7	3071.0	-1.59	.10	-.15
DRM	23	4	84.5	24	4	94.8	3189.0	-1.23	.22	-.11
PFM	18	4	87.5	18.4	5	93.4	3364.5	-0.69	.48	-.06
CDA	13	2	80.2	14	5	96.8	2942.5	-1.97	.04	-.18
PFP	7	2	87.4	7	2	93.4	3357.5	-0.72	.46	-.07
Marital Status										
Married (n = 108)				Single/Divorced (n = 74)			U	Z	p	r _{rb}
Scale	Median	IQR	Mean R.	Median	IQR	Mean R.				
SOGS	0	0	88.3	0	1	96.2	3650.0	-0.99	.20	-.09
FL Total	108	16.3	93.8	107	21.3	88.1	3744.0	-0.72	.47	.06
FF	30	6	95.9	28.5	5	85.0	3516.0	-1.37	.17	.12
IIF	16	2	90.4	16	3	93.1	3877.5	-0.34	.73	-.03
DRM	24	3	94.3	23	5	87.4	3690.5	-0.88	.38	.08
PFM	18.4	3.3	95.1	18	5	86.3	3608.0	-1.11	.26	0.9
CDA	13.1	3.3	88.3	13.1	4.8	96.2	3650.0	-0.99	.31	-.09
PFP	7	2	93.2	7	2	88.9	3807.5	-0.54	.58	.05
IPS Participation										
Yes (n = 52)				No (n = 130)			U	Z	p	r _{rb}
Scale	Median	IQR	Mean R.	Median	IQR	Mean R.				
SOGS	0	1	91.4	0	1	91.6	3373.0	-0.02	.98	-.00
FL Total	107	19.3	85.3	108	15.9	94.0	3059.0	-1.00	.32	-.10
FF	28	5.2	82.7	29	6	95.0	2920.0	-1.43	.15	-.14
IIF	16.3	2.8	89.2	16	2.4	92.4	3259.0	-0.38	.70	-.04
DRM	24	3.3	90.3	24	4.8	92.0	3320.0	-0.19	.85	-.02
PFM	18	4.3	85.7	18.4	5	93.8	3078.5	-0.94	.35	-.09
CDA	13	2.3	82.8	14	4	95.0	2927.0	-1.41	.15	-.13
PFP	7	2	90.5	7	2	91.9	3329.0	-0.16	.87	-.02
Financial Literacy Training										
Yes (n = 18)				No (n = 164)			U	Z	p	r _{rb}
Scale	Median	IQR	Mean R.	Median	IQR	Mean R.				
SOGS	0	1	95.6	0	1	91.0	1402.0	-0.35	.66	.05
FL Total	118	25	112.7	107.0	15.3	89.2	1094.0	-1.80	.07	.26
FF	32.5	7	108.2	29	6	89.7	1175.0	-1.42	.16	.20
IIF	17.5	2.8	111.2	16	2	89.3	1121.5	-1.67	.09	.24
DRM	25	7.3	108.0	24	4	89.7	1179.5	-1.40	.16	.20
PFM	20	3	106.9	18	4	89.8	1198.0	-1.31	.19	.19
CDA	15	6.3	109.9	13.0	4	89.5	1144.0	-1.56	.11	.22
PFP	7	2	91.8	7	2	91.5	1471.5	-0.02	.98	.00

Note. Nonparametric Mann–Whitney U tests are reported for all outcomes. r_{rb} values are Glass rank–biserial correlations. SD = standard deviation; IQR = Q3 – Q1. SOGS = South Oaks Gambling Screen, FL Total = Financial Literacy total score, FF = Fundamentals of Finance, IIF = Investment Instruments and Financing, DRM = Debt and Risk Management, PFM = Personal Finance Management, CDA = Cryptocurrencies and Digital Assets, and PFP = Personal Financial Planning.

In the multi-group analyses (Kruskal–Wallis with Bonferroni-adjusted Dunn tests), significant differences were observed in several outcomes based on education levels: SOGS ($H(2) = 19.02$, $p < .001$; $\epsilon^2 = 0.095$; lower/secondary > bachelor and > postgraduate), FL Total ($H(2) = 11.23$, $p = .0036$; $\epsilon^2 = 0.052$; lower/secondary < bachelor), FF ($H(2) = 20.58$, $p < .001$; $\epsilon^2 = 0.104$; lower/secondary < bachelor and < postgraduate), PFM ($H(2)$

= 7.85, $p = .0197$; $\varepsilon^2 = 0.033$; lower/secondary < bachelor), and CDA ($H(2) = 11.57$, $p = .003$; $\varepsilon^2 = 0.054$; lower/secondary < bachelor). Income also stratified outcomes: SOGS ($H(2) = 9.02$, $p = .011$; $\varepsilon^2 = 0.039$; low > middle), FL Total ($H(2) = 7.00$, $p = .030$; $\varepsilon^2 = 0.028$; low < high), DRM ($H(2) = 6.41$, $p = .040$; $\varepsilon^2 = 0.025$; high > low), and CDA ($H(2) = 6.76$, $p = .034$; $\varepsilon^2 = 0.027$; high > low, borderline at $p = .050$). Age differences were confined to CDA ($H(3) = 11.17$, $p = .011$; $\varepsilon^2 = 0.046$), where the 20–29 group exceeded 40–49 ($p = .019$; the 20–29 vs 30–39 contrast was marginal at $p = .051$). Savings rate showed an omnibus effect only for PFM ($H(4) = 9.78$, $p = .044$; $\varepsilon^2 = 0.033$), but no pairwise comparisons survived multiplicity correction. Kruskal–Wallis statistics and Bonferroni-adjusted Dunn post-hoc comparisons are reported in Table 6.

Table 6.

Multi-group comparisons (Kruskal–Wallis) and dunn post-hoc pairwise comparisons (Bonferroni-adjusted): Age, education level, income level, savings rate ($N = 182$)

Age (n)												
¹ 20-29 (57)			² 30-39 (51)		³ 40-49 (51)		⁴ 50+ (23)		χ^2 (df = 3)	ϵ^2	Dunn Post-hoc	
Scale	Median (IQR)	Mean Rank	Median (IQR)	Mean Rank	Median (IQR)	Mean Rank	Median (IQR)	Mean Rank				
SOGS	0 (1)	101.9	0 (0.5)	88.8	0 (0)	83.9	0 (0)	88.7	5.74	0.02	3 < 1	
FL Total	109 (19)	105.1	106 (14.5)	81.8	105 (14)	83.6	112 (18)	97.1	6.95	0.02		
FF	30 (6)	94	29 (5)	91.3	28 (5.5)	84.8	30 (7)	100.7	1.69	0.00		
IIF	16 (3)	90.4	16 (2)	85.4	16 (2)	90.3	17 (3.5)	110.4	3.82	0.00		
DRM	25 (5)	103.4	23 (4)	76.6	24 (3)	92.9	24 (3.5)	92.1	7.10	0.02		
PFM	19 (4)	101.3	18 (5)	81.7	18 (4)	87.8	20 (3.5)	97.2	4.30	0.01		
CDA	14 (6)	110.0	12 (3.5)	83.2	12 (2.5)	80.3	14 (2.5)	88.9	11.2*	0.05		
PFP	7.16 (1)	102.8	7 (2)	92.4	7 (2)	79.8	7 (2)	87.6	5.52	0.01		
Education Level (n)												
¹ Primary/ Secondary (32)			² Bachelor's (109)		³ Postgraduate (41)				χ^2 (df = 2)	ϵ^2	Dunn Post-hoc	
Scale	Median (IQR)	Mean Rank	Median (IQR)	Mean Rank	Median (IQR)	Mean Rank						
SOGS	1 (2)	120.1	0 (0)	86.2	0 (0)	83.2			19.02**	0.09	3 < 2 < 1	
FL Total	100 (11.7)	63.3	108 (17)	98.4	110 (18)	95.3			11.23**	0.05	1 < 2	
FF	27 (2.1)	54.3	30 (6)	96.7	30 (7)	106.8			20.58**	0.10	1 < 2 < 3	
IIF	15 (2.1)	88.2	16 (2.4)	89.1	16.6 (3)	100.5			1.60	0.00		
DRM	22.5 (3.0)	74.3	24 (4)	97.5	24 (5)	88.9			5.00	0.02		
PFM	17 (3.6)	67.9	19 (4)	97.1	19 (5)	95.1			7.85*	0.03	1 < 2	
CDA	12 (1.1)	64.2	14 (4)	99.4	14 (4)	91.7			11.57**	0.05	1 < 2	
PFP	7 (1.4)	82.6	7 (2)	98.7	7 (2)	79.3			5.37	0.02		
Income Level (n)												
¹ Low (68)			² Middle (102)		³ High (12)				χ^2 (df = 2)	ϵ^2	Dunn Post-hoc	
Scale	Median (IQR)	Mean Rank	Median (IQR)	Mean Rank	Median (IQR)	Mean Rank						
SOGS	0 (1)	102.7	0 (0)	86.1	0 (0)	73.8			9.02*	0.04	2 < 1	
FL Total	107 (16.5)	82.7	108 (17)	93.4	110 (20)	125.3			7.00*	0.03	1 < 3	
FF	28.5 (5)	85.9	29 (6)	91.2	32.5 (9.5)	126.1			5.99	0.02		
IIF	15 (2.1)	84.6	16 (2)	93.7	17.2 (4.3)	111.8			3.21	0.01		
DRM	23 (4)	81.6	24 (4)	94.8	25 (2.8)	120.1			6.41*	0.02	1 < 3	
PFM	18 (3.3)	82.6	19 (5)	96.6	18.7 (2.5)	98.3			3.14	0.01		
CDA	13 (3)	80.8	14 (4)	95.5	15 (2.73)	118.4			6.76*	0.03	1 < 3	
PFP	7 (2)	89.5	7 (2)	90.7	7.58 (1)	109.6			1.61	0.00		
Savings Rate (n)												
None (46)			0%-5% (47)		6%-9% (24)		10%-14% (18)		15%-19% (47)		χ^2 (df = 4)	ϵ^2
Scale	Median (IQR)	Mean Rank	Median (IQR)	Mean Rank	Median (IQR)	Mean Rank	Median (IQR)	Mean Rank	Median (IQR)	Mean Rank		
SOGS	0 (1)	97.8	0 (0)	85.7	0 (0)	81.2	0 (1)	101.7	0 (1)	92.5	4.68	0.00
FL Total	107 (19.3)	86.8	107 (16)	86.3	104 (7.8)	78.5	108 (26)	91.2	110 (21)	108.0	6.91	0.02
FF	28.5 (6)	86.6	30 (6.5)	95.9	28 (3.9)	75.9	28 (5.4)	84.6	31 (6.6)	102.6	5.26	0.00
IIF	16 (2)	89.8	15 (2)	83.9	15 (3)	78.1	16.3 (4.4)	89.5	17 (4)	108.4	7.64	0.02
DRM	24 (4)	89.3	23 (5)	86.2	23.4 (2.3)	88.7	23.5 (5.8)	82.4	25 (3)	103.9	3.82	0.00
PFM	17 (5)	80.5	18 (3.5)	86.1	18 (4.3)	79.7	19.5 (2.6)	107	19 (5)	107.7	9.78*	0.03
CDA	13 (3)	87.6	13 (3.5)	89.3	13.0 (2)	83.5	13.1 (4)	88.8	14 (5)	102.6	3.16	0.00
PFP	7 (2)	88.8	7 (2)	79.3	7 (1)	95.1	8 (2.2)	107.9	7.16 (2)	98.2	5.48	0.00

Note. Kruskal–Wallis tests summarizing significant multi-group effects. * $p < .05$, ** $p < .001$. Effect size computed as ϵ^2 . Dunn's test with Bonferroni adjustment following significant Kruskal–Wallis. Savings rate was significant only for PFM, but no pairwise comparisons survived multiplicity correction.

Discussion

This study examined the relationship between financial literacy and gambling behaviour. Descriptive distributions provide explanatory information about sample characteristics. Analyses conducted on sociodemographic variables have highlighted various findings: Two-group comparisons showed a significant difference in favour of males in CDA, while multiple group comparisons drew attention to the importance of the relationships between income and education level with SOGS and FL. Finally, the correlation analysis showed only a negative significant relationship between FF and SOGS; FL Total and SOGS yielded a borderline p-value. On the other hand, no significant relationship was found between financial literacy's IIF, DRM, PFM, CDA, and PFP sub-dimensions and gambling.

Descriptive statistics reveal important findings regarding the frequency and types of gambling behaviour within the sample studied. It was determined that 49 participants gambled, with the maximum amount wagered in a single day ranging from £5 to £1,000,000. Forty-one per cent of participants had experienced at least one type of gambling, with stock market transactions, the National Lottery, and Sports Toto/Lotto emerging as the most common types of gambling. The fact that these types are legal in Türkiye and more socially acceptable may explain why they are more commonly experienced than other illegal types. Indeed, it is known that legally and socially relatively acceptable gambling activities are more widespread ([Barone & Graffigna, 2025](#)). The DSM-5 emphasises that the development of gambling disorder follows a progressive course over time, with individuals initially engaging in a few types of gambling but potentially increasing the number of gambling activities over time ([American Psychiatric Association, 2022](#)). [Blażczynski and Nower's \(2002\)](#) Pathways Model, which suggests that individuals become pathological gamblers through different pathways, emphasises that people who have access to gambling and have experienced it may, over time, meet the criteria for pathological gambling through behavioural conditioning and cognitive distortions. This information from the literature on the development process of gambling disorder is a noteworthy finding in terms of the potential effects of the ratio obtained from the research. On the other hand, the pathological gambler ratio obtained from our research sample is 2.7%. It is known that the group that is considered at risk but not pathological gamblers, which represents a broader group rather than the small number of pathological gamblers representing severe cases, contributes more to the social harm caused by gambling ([Volberg et al., 2021](#)). When the gambling experience rate and the pathological gambler rate obtained from the research sample are considered together, this may be an important finding in terms of the design of prevention efforts targeting the potential social harms of gambling. The results at the correlation level indicate that the relationship between SOGS and FL Total is negative but insignificant (borderline). Only a negative and significant relationship was found between FF and SOGS. These findings are important in that they show that not all dimensions of financial literacy are related to gambling behaviour in the same way. This finding from our research is consistent with the literature. According to [Barone and Graffigna's \(2025\)](#) meta-analysis, studies show that there may be generally negative relationships between financial literacy and gambling, but that the relationship may vary depending on gambling access conditions and FL sub-dimensions. The existence of a negative relationship between the FF sub-dimension and SOGS suggests that basic financial knowledge (such as debt management and budgeting), rather than complex and detailed financial information, has dynamics that are opposite to gambling behaviour. We believe that individuals with high FF may be

inclined to view gambling as an expenditure from a budget allocated for entertainment, rather than as an investment tool or a means of making money quickly. Considering people's tendency to prefer small, immediate rewards over larger, delayed rewards ([Frederick et al., 2003](#)), it can be said that gambling facilitates the pursuit of instant gratification, while basic financial literacy may play a countervailing role in encouraging the opposite behaviour. FF may contribute to making individuals more resilient to the promise of instant rewards and the uncertainty associated with the risk of loss in gambling, as it is thought to encourage them towards delayed reward-oriented behaviours such as saving. Contrary to the tendency inherent in gambling to chase losses ([Kahneman & Tversky, 1979](#)), we believe that individuals with a high level of basic financial knowledge would prefer to avoid further financial losses rather than take greater risks to recoup their losses. This perspective may explain the negative relationship we found in our research. However, experimental/longitudinal studies are needed to shed light on the direction and nature of the relationship. [Watanapongvanich et al. \(2022\)](#) reported that, in US data, financial literacy can have a protective function in areas with limited access to gambling. Considering nationwide regulations in Türkiye, studies testing how accessible and prevalent types of gambling intersect with FL dimensions would be particularly valuable for local policy design.

Level of education shows significant differences in both gambling and financial literacy scores. Those in the lower education group had higher SOGS scores and lower FL Total, FF, PFM and CDA scores. A similar pattern was observed for income level; those in the low-income group had higher SOGS scores, lower FL Total scores, and differences in favour of high income were observed in some sub-dimensions (DRM, CDA). A significant difference for the age variable was only observed in the CDA sub-dimension. Those aged 20–29 were found to have higher CDA scores than those aged 40–49. The literature indicates that low education and low income levels show similar results in terms of both financial literacy ([Hamurcu et al., 2025](#)) and more frequent gambling ([Calado & Griffiths, 2016](#); [Moreira et al., 2023](#)).

In the two-group analyses, differences based on gender were only observed in the CDA subscale, with men scoring higher than women. No significant gender-based differences were observed in the SOGS, FL Total, or other subscales. No significant differences were found in marital status, IPS participation, or prior financial literacy training. These results suggest that gender-based differences may exist in interactions with digital assets, but that binary distinctions are not determinative for overall gambling severity and financial literacy total and sub-dimensions.

The differentiation of the FF sub-dimension of SOGS from FL Total suggest that content focusing on fundamental financial behaviours and principles (e.g., budgeting, debt risk management, spending and saving planning) could be considered more functional against the instant reward cycle of gambling. The systematic correlation of educational and income inequalities with both FL and SOGS highlights the importance of considering the consideration of targeted educational programmes (particularly for low-education/low-income groups) and access-focused regulations (e.g., marketing and visibility of digital assets).

Limitations and Suggestions for Future Research

As this study has a cross-sectional design, it is not possible to determine causality. The sample size (N=182) may be too small to detect small effects. As the research data is self-reported, socially acceptable responses are possible. Non-parametric methods were preferred as the clustering of SOGS scores around zero and the skewed distribution may affect statistical sensitivity. The high proportion of males in the sample may indicate possible gender imbalances. The sample is characterized by high educational attainment (82.4% bachelor's degree or higher) and universal internet banking usage, representing a specific demographic profile. As lower socioeconomic status is a recognized risk factor for gambling, the limited representation of such groups implies that the findings may not fully reflect dynamics in high-risk populations. Therefore, results should be interpreted within the context of an educated, digitally literate sample and generalized with caution to lower socioeconomic segments.

Longitudinal and/or experimental studies (e.g., financial education interventions) may shed light on the direction of the relationship and its mechanisms. Larger samples with stratified sampling, ensuring adequate representation of low-education/low-income groups, may be preferable. Gambling access and exposure indicators (physical digital access, advertising/visibility, etc.) can be measured to address interactions with FL sub-dimensions. Behavioural determinants (e.g., impulsivity, delayed reward preference, risk attitude) and financial decision components can be addressed in the same model for advanced analysis. Randomised controlled trials evaluating the impact of sub-dimension-focused educational content (particularly FF) can provide high value in terms of policy and practice.

This study demonstrated that, rather than all dimensions of financial literacy, the Fundamentals of Finance (FF) dimension in particular revealed a negative relationship with gambling intensity, and that education and income levels created systematic differentiation in both financial literacy and gambling indicators. The findings suggest that targeted educational programmes focusing on basic financial behaviours and principles (especially for low education/low income groups) may be promising in mitigating gambling-related risks. However, due to the cross-sectional design, cautious interpretation is warranted without establishing causality, and the evidence base should be strengthened through longitudinal/experimental studies testing the direction and mechanisms of the relationship.

Ethic

Ethical approval for this study was obtained from Necmettin Erbakan University ethics committee (Decision No: 2025/5575, Date: 21.02.2025).

Author Contributions

Conception: Ö. Karadağ Ak, M. Ak; Design: A. Hazar, Ş. Babuşcu; Supervision: M. Ak, A. Hazar, Ş. Babuşcu; Data Collection: Ö. Karadağ Ak, A. Akyıldız, A. E. Yazgan; Analysis: A. Akyıldız, A. E. Yazgan; Literature Review: A. Akyıldız; Writing: A. Akyıldız, Ö. Karadağ Ak; Critical Review: M. Ak, A. Hazar, Ş. Babuşcu.

Conflict of Interest

The authors declare that there is no conflict of interest regarding this study.

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